

SANJANA REDDY PATLOLLA

Phone: +1 (551) 235 4378 Email: sanjanareddy2901@gmail.com [LinkedIn](#) Location: Edison, New Jersey

Summary

AI/ML and Software Engineer skilled in Python, R, C++, SQL, JavaScript, with hands-on experience in ML frameworks such as TensorFlow, PyTorch, XGBoost and cloud platforms. Worked on regulatory compliance, AI governance, data pipeline automation, financial risk modeling, scalable ML solutions, and system optimization.

Skills

Programming Languages: Python, R, C++, JavaScript, SQL/DBMS

Data and Machine Learning: XGBoost, Stacked LSTM, SBERT, PostgreSQL, Oracle SQL, MongoDB, Pandas, NumPy, Scipy, TensorFlow, PyTorch, Keras, Scikit-learn, NLP, OpenCV, Data Collation

Tools and Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI, Agile Methodologies, Git/GitHub

Web and Cloud: HTML5, CSS3, React.js, Node.js, Angular.js, Node-Red, Tinkercad, REST APIs, Django, Kubernetes, Linux, Flask

Soft Skills: Problem Solving and Analytical Thinking, Collaboration, Oral and Written Communication

Work Experience

Research Assistant - Stevens Institute of Technology Feb 2025 - Present, New Jersey

- Spearheaded AI-driven compliance solutions, optimizing LLMs for financial regulations and risk assessment.
- Implemented an SBERT-based query system to enhance compliance information retrieval efficiency.

ML Research Assistant - VIT University Sept 2022 - May 2023, India

- Developed XGBoost heart disease model with 99% accuracy on 40K+ ECGs, classifying 7 conditions in 30ms.
- Optimized AI diagnostics, improving speed and precision by 85% with scalable ML-driven feature extraction.
- Designed automated data pipelines for ECG preprocessing, ensuring seamless model training classification.

Data Analyst Intern - LetsGrowMore Jul 2022 - Aug 2022, India

- Created Python scripts for image transformation (grayscale, inversion, blurring), improving accuracy by 40%.
- Developed stock market and sales forecasting models using Stacked LSTM, improving decision-making.

Software Engineer Intern - Abbeysoft IT Services Nov 2021 - May 2022, India

- Optimized web interfaces with HTML, CSS, JavaScript, and AngularJS, boosting user satisfaction by 30%.
- Engineered responsive UI components, improving performance, speeding page load, and increasing Interaction.

Artificial Intelligence Training Engineer - SmartInternz Jun 2021 - Jul 2021, India

- Constructed an AI-powered species identification web app using IBM Watson Studio, achieving 92% accuracy.
- Integrated deep learning models with Flask, TensorFlow, and Keras, reducing processing time by 35%.

Academic Projects

Financial Web Application

- Developed a financial web app using Python (Django/Flask) and JavaScript (React/Angular) to analyze bank-SHG linkage models, manage financial data, and assess loan risks.

IoT-Based Farmers' Helper

- IoT-Based Farmers' Helper enhanced smart farming with sensor-driven irrigation, weather prediction (R programming), and automated crop protection (Node-RED, Tinkercad) for efficiency and sustainability.

Research Publications

A Machine Learning Based Approach to Early Stage Diabetes Prediction

- Implemented an AI-driven early diabetes prediction model with high accuracy, presented at the 6th International IEEE Conference on Electronics, Communication, and Aerospace Technology.

An Intelligent Flood Automation System Using IoT and Machine Learning

- Developed an IoT-based flood monitoring and early warning system using machine learning, showcased in IOS Press International.

Education

Stevens Institute of Technology Sept 2023 - May 2025

Master of Science in Information Systems

CGPA 3.94

Vellore Institute of Technology

Jun 2019 - May 2023

Bachelor of Technology in Computer Science and Engineering